

Exam Date & Time: 22-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

Digital Design Verification [ECE-4062]

Marks: 50

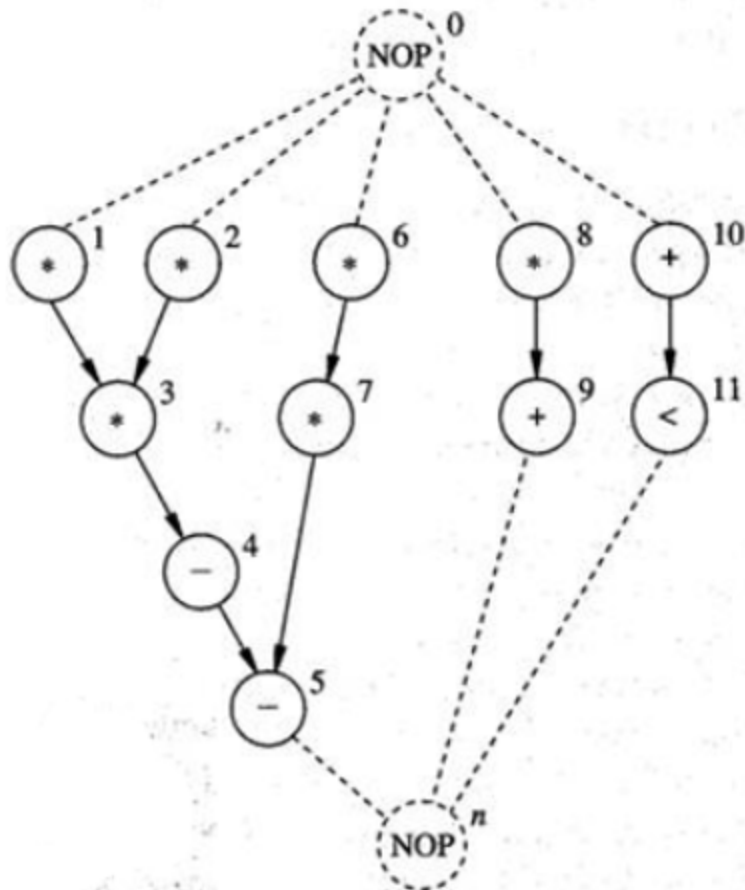
Duration: 180 mins.

PART-A

Answer all the questions.

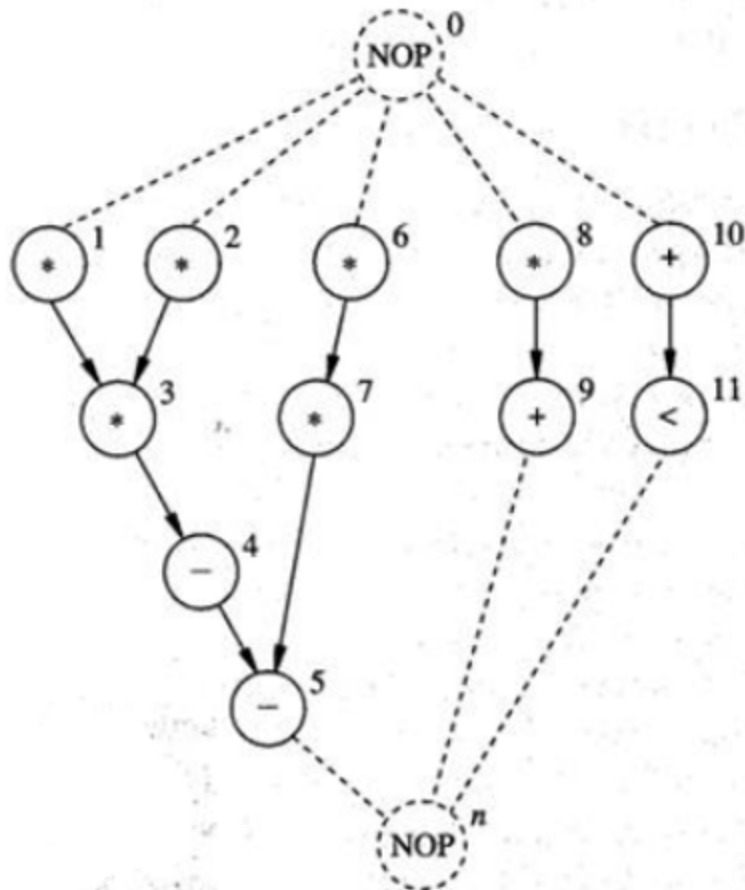
- 1) Draw ROBDD for 3 bit synchronous even counter. Show all the steps starting from BDD. Perform ITE algorithm for ROBDD. (4)
 - A)
 - B) Draw Control and data flow graph for the following sequential statements $X=A*B$, $Y=C+D$, if $X < Y$ then $Z=Y-X$ else $Z=X-Y$. (3)
 - C) Explain the flow of digital design flow and verification (3)
- 2) (4)
 - A)

Apply ASAP and ALAP algorithm for the given data flow graph shown below. Determine the latency using ASAP and mobility using ALAP. Draw the scheduled graph. Assume all the operations have unit execution delay.



- B) Define latency, mobility and resource constraints (3)
- C) Perform ITE algorithm for the expression
 $\text{sum} = A \text{ XOR } B \text{ XOR } C$ (3)
- 3) Apply Hu algorithm for the given expression $F = [(C + (A * B)) - ((B * C) + A)] < [P + Q] + Y$
 Assume $\gamma = 4$, $P(0) = 1, P(1) = 3, P(2) = 4, P(3) = 2, P(4) = 2$. Determine the resource constraints. Draw the rescheduled graph with resource constraints (4)
- A) (3)
- B) (3)

Determine minimum latency under resource constraint for the following sequencing graph using LIST-L scheduling algorithm. Draw the scheduled graph. Assume mobility $m_1=m_2=m_3=m_4=m_5=0$, $m_6=m_7=1$, $m_8=m_9=m_{10}=m_{11}=2$.



- C) Write the formula to calculate the minimum number of resources using Hu Algorithm and Explain. (3)
- 4) Find the essential prime implicant for the set of prime implicants $F=\{C_1, C_2, C_3, C_4\}$ where (4)
- A) $C_1=x_1'x_3'x_4'$ $C_2=x_1x_2$ $C_3=x_3x_4$ $C_4=x_2x_3'x_4'$ using ESPRESSO algorithm. (4)
- B) Using , Quine Mcculuskey method, obtain prime implicants and minimal expression for the following function $F=\sum m(6,7,8,9)+d(10,11,12,13,14,15)$ (3)
- C) Determine the prime implicant for the following function using iterated consensus method (3)
- $$F(A,B,C) = A'B'C'+A'BC'+A'BC+AB'C'+AB'C+ABC'$$
- 5) Write a system verilog code for Full adder (4)

- A)
- B) Write a system verilog code and test bench for buffer (3)
- C) Write the syntax for while loop and do while loop in system verilog with an example each (3)

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